INCIDENT REPORT · NIST SP 800-61

INC-0419 · Suspected guest-to-host escape via the vGPU path

Opened Nov 8, 02:14 CST · Incident Commander RK · Phase: Eradication

1 Identification

Incident ID	INC-0419	Severity	Critical
Category	Unauthorized access · guest-to-host escape		
Detected	02:14:07 CST · Sentry · VMM audit: unexpected ioctl 0xC0186441 on vm-7f3a		
Functional impact	Medium · FIN-02 locked. Data center back to normal with 1 host out of service		
Information impact	Privacy breach suspected · billing DB access under review, nothing confirmed taken		
Recoverability	Supplemented · reimage host, patch the driver, restore FIN-02 from backup		
Incident Commander	RK · Incident Commander	Current phase	**Eradication**
Systems	b14-gpu-07 (Hall B · rack 14) · vm-7f3a · agent-7f3		

2 Response phases

✓ Detection and analysis	✓ Containment	3 Eradication	4 Recovery	5 Post-incident
Confirm it is real, scope it, rate it.	Stop it spreading. Preserve evidence.	Remove the cause: patch, reimage, rotate.	Return systems to service and watch them.	Lessons learned and control changes.
Done · 02:14:08 to 02:14:09	Done · 02:14:09 to 02:14:17	Current · since 02:14:17	Not started	Not started
Set by Sentry	Set by Warden	Set by RK		

3 Timeline

02:14:07 T+0.0s	**Sentry**	VMM audit: unexpected ioctl 0xC0186441 on vm-7f3a
02:14:07 T+0.3s	**Sentry**	IOMMU fault: DMA write outside guest range
02:14:07 T+0.5s	**Sentry**	host EDR: qemu spawned /bin/sh
02:14:08 T+1.0s	**Sentry**	Correlated 6 signals, opened INC-0419

02:14:08 T+1.0s	Sentry	**Phase set to Detection and analysis**	
02:14:08 T+1.8s	Triage	Attack graph: 1 suspect host, 11 co-resident guests, 2 exposed tokens, mgmt plane reachable	
02:14:09 T+2.0s	Warden	**Phase set to Containment**	
02:14:09 T+2.4s	Warden	Isolated network: detached vNIC of vm-7f3a	
02:14:09 T+2.9s	Warden	Revoked workload credentials (2 tokens)	
02:14:10 T+3.3s	Warden	Cordoned host b14-gpu-07	
02:14:10 T+3.5s	AL	Corp IT reported ransomware on FIN-02. Tracked as a parallel event	
02:14:10 T+3.7s	Triage	Billing database read by a corp service account in the window. Marked Accessed	
02:14:10 T+3.8s	Warden	Paused guest vm-7f3a	
02:14:11 T+4.6s	Warden	Captured guest memory via trusted management path	
02:14:12 T+5.3s	Warden	Failed over 11 co-resident guests to clean hosts	
02:14:12 T+5.5s	Warden	Requested approval: quarantine host b14-gpu-07	
02:14:14 T+7.4s	RK	Approved quarantine with ⌘↵	
02:14:16 T+9.4s	Warden	Quarantined host b14-gpu-07. Incident contained	
02:14:17 T+10.4s	RK	**Phase set to Eradication**	
02:14:19 T+12.4s	Scribe	Filed report, synced to 6 operators	

4 Actions and authorization

TIME	ACTION	TARGET	TAKEN BY	AUTHORIZED BY		REVERSIBLE
02:14:09	**Isolate network** Cut the guest's network. Reversible.	vm-7f3a	Warden	Playbook guest-escape v3	automatic, policy-bounded	Yes
02:14:09	**Revoke credentials** 2 tokens used outside scope. Reissue is possible.	agent-7f3	Warden	Playbook guest-escape v3	automatic, policy-bounded	Yes
02:14:10	**Cordon host** No new placements while the host is untrusted.	b14-gpu-07	Warden	Playbook guest-escape v3	automatic, policy-bounded	Yes

TIME	ACTION	TARGET	TAKEN BY	AUTHORIZED BY	REVERSIBLE
02:14:10	**Pause guest** Freeze vCPUs so the memory can be captured.	vm-7f3a	Warden	Playbook guest-escape v3 automatic, policy-bounded	Yes
02:14:11	**Capture memory** Read-only capture over the trusted management path.	vm-7f3a	Warden	Playbook guest-escape v3 automatic, policy-bounded	Yes
02:14:12	**Failover guests** Move neighbours to clean hosts.	11 co-resident guests	Warden	Playbook guest-escape v3 automatic, policy-bounded	Yes
02:14:12	**Quarantine host** qemu on this host spawned a shell after a DMA fault.	b14-gpu-07	Warden	RK · Incident Commander human approval (you)	Yes

5 Scope of compromised data

ASSET	SYSTEM	CLASS	RECORDS AT RISK	STATUS
Corporate file share FIN-02 IT confirms the 01:00 offline backup is clean. Restore starts once the file server is rebuilt.	Corp IT · file server NAS-02	Confidential	184K locked	Locked (ransomware) AL · 02:14:17
Workload tokens Both tokens revoked by Warden at T+2.9 s. Still marked suspected until the flow logs are read. New tokens issue only after review.	DFW-2 · workload IAM	Restricted	0 to 2 not known yet	Exfil suspected Warden · 02:14:09
Customer billing database Query logs pulled: 3 large SELECTs from 10.20.4.17 (corp IT). Reviewing for exfiltration.	Corp IT · billing Postgres	Restricted	0 to 2.4M not known yet	Accessed Triage · 02:14:15
Guest memory image vm-7f3a, 64 GiB, captured over the trusted management path. sha256 9f3c…e1a0. Read-only, off the network.	DFW-2 · evidence vault	Restricted	1 held	Evidence preserved Warden · 02:14:11
Tenant model weights Snapshot taken with the memory capture. 142 shards hashed and held in the evidence vault.	DFW-2 · tenant vm-7f3a	Restricted	142 held	Evidence preserved Scribe · 02:14:11
Co-resident tenant volumes All 11 failed over to clean hosts. No cross-tenant access seen.	DFW-2 · b14-gpu-07	Confidential	0 of 11	Intact Warden · 02:14:12
Management-plane credentials VLAN 40 pruned from the host at quarantine. No sign of use.	DFW-2 · BMC / IPMI, VLAN 40	Restricted	0 of 14	Intact Warden · 02:14:15

6 Evidence and chain of custody

vm-7f3a.mem
64 GiB · Warden · 02:14:11

Hash verified

SHA-256 `9f3c92ff5bc41ac3b4436d5ebe96f6d428aefefc9eb79debb1f27ebe4367e1a0`

Source Guest memory of vm-7f3a, over the trusted management path

- 02:14:11 **Captured, read-only** — Warden
- 02:14:11 **Hashed, SHA-256 recorded** — Scribe
- 02:14:11 **Stored, off the network, write-once** — Evidence vault
- 02:14:18 **Custody accepted (Incident Commander)** — RK

qemu-2231.core
1.2 GiB · Warden · 02:14:12

Hash verified

SHA-256 `41be064ff1c3715de0b49949a851ca52c495d5dd26ea296aebf01955154407d2`

Source Core dump of the qemu process on b14-gpu-07

- 02:14:12 **Captured from the paused host process** — Warden
- 02:14:12 **Hashed, SHA-256 recorded** — Scribe
- 02:14:12 **Stored, off the network, write-once** — Evidence vault
- 02:14:18 **Custody accepted (Incident Commander)** — RK

b14-gpu-07-flows.pcap
38 MB · Sentry · 02:14:12

Hash verified

SHA-256 `c0a7679a2de131f8ecb44c42000e98bd728f328a26edd6cf5add912a7e195f19`

Source East-west flow capture for b14-gpu-07, 02:14:07 onward

- 02:14:12 **Exported from the flow recorder** — Sentry
- 02:14:12 **Hashed, SHA-256 recorded** — Scribe
- 02:14:12 **Stored, off the network, write-once** — Evidence vault
- 02:14:18 **Custody accepted (Incident Commander)** — RK

7 Communications and decisions

02:14:09	Six reversible containment steps authorized by playbook guest-escape v3	Policy
02:14:10	Incident Commander assumed command	RK
02:14:10	FIN-02 ransomware is tracked in this war room as a parallel event	AL
02:14:14	Quarantine b14-gpu-07 approved by RK	RK

#command 13 messages **#legal** 4 messages **#exec** 4 messages **#comms-pr** 3 messages **#vendors** 3 messages

Notification clocks have not started. They start only when counsel decides materiality (example obligations: SEC 8-K Item 1.05 within 4 business days; GDPR within 72 hours).

8 Lessons learned

Completed at the post-incident review, within 14 days of recovery. Owner: RK.

- [ ] What detected it, and how fast? Did the signal come from the right place?
- [ ] What slowed containment, and what would have sped it up?
- [ ] Which playbook step, policy or control should change?
- [ ] Who needed to know sooner, and how did they find out?
- [ ] Which of the open questions in the brief were answered, and how?

Written by Scribe. Entries are appended and attributed, never edited. Evidence hashes are SHA-256.